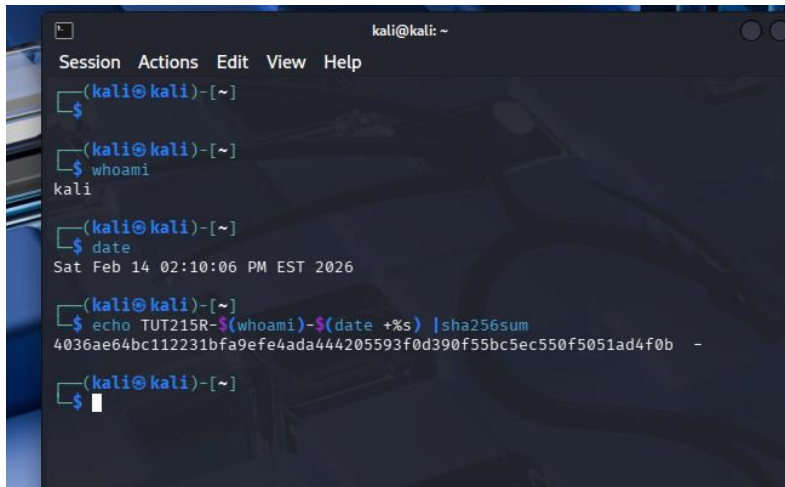


STUDENT NUMBER :222932050

SURNAME AND INITIALS :Mahlangu S

COMPUTER SECURITY

PART3 -PERSONAL LAB TOKEN



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$  
(kali@kali)-[~]  
$ whoami  
kali  
(kali@kali)-[~]  
$ date  
Sat Feb 14 02:10:06 PM EST 2026  
(kali@kali)-[~]  
$ echo TUT215R-$(whoami)-$(date +%s) |sha256sum  
4036ae64bc112231bfa9efe4ada444205593f0d390f55bc5ec550f5051ad4f0b -  
(kali@kali)-[~]  
$
```

PART4 -CHAPTER1 TASKS

TASK 1 -RULES OF ENGAGEMENT

* SCOPE

-IPV4 Prefix : 192.168.50.0/24

The testing is to be limited to this network only which has ip format of 192.168.50.0/24.The ip address of the device where the test will take place is 192.168.50.2.

-Timeframe :This test will be effective from the 13 February 2026 until 23 February 2026.

-Tools: Only footprinting tools will be used, such as search engines ,WHOIS services ,DNS lookuptools and network mapping tools .

-Data access : Every device within this range 192.168.50.0-192.168.50.254 can be accessed during the testing.

* ALLOWED ACTIVITIES (Footprinting only)

-The ethical hacker is only permitted to use footprinting only this includes when gathering information this includes both active and passive footprinting.

* Prohibited activities (Exploitation)

-The ethical hacker is not allowed to use tools which they are not given permission to use .Only footprinting tools and methods ,are permitted to be used in this test.

-The ethical hacker is not permitted to obtain any of the data in an illegal way , everything should be done in a professional and legal manner . Failure to do so will result in the ethical hacker facing arrest charges .

-No password attacks can be performed .

* CONFIDENTIALITY CLAUSE

-The information that is discovered by the ethical hacker is not to be discussed with third parties.

-After discovering the weaknesses of the organization , the ethical hacker must cover all their tracks by implementing the security procedures .

-The client must provide all the information that will be necessary to the ethical hacker .

-If a security breach occurs while conducting the test , the ethical hacker must inform the client immediately.

-Failure to follow all these rules will result in the hacker being sued and facing jail time.

TASK 2-ETHICAL SCENARIO ANALYSIS

* Scanning without permission

-”Network scanning, particularly when performed without proper authorization, can infringe on the privacy of individuals and organizations. Ethical hackers and cybersecurity professionals must ensure that they only scan networks they own or have explicit permission to assess.”(İbrahim Korucuoğlu,2025)

- WHOIS lookup

Authorization: Anyone who uses WHOIS lookup should ensure that they have permission from the organization that they are looking for because if you do not have the permission, then you could face legal charges.

Confidentiality: The information that is discovered from WHOIS lookup should not be revealed to anyone without the permission of the organization, and this is done to protect the organization.

Scope: The WHOIS lookup should be used for legal purposes such as doing research and should not be used for illegal and malicious purposes.

- Metadata extraction

Authorization: You can only use metadata extraction if you have the permission of the owner of the company or users.

Confidentiality: The information discovered should not be revealed to anyone because it is sensitive data.

Scope Control: When doing metadata extraction, one must make sure that they only extract what is defined in the scope because when you extract more, it can often lead to legal issues.

- Brute-force attempt

Authorization: It is legal to do a brute-force attempt if an ethical hacker has the permission of the individual or an organization that has hired them to improve their security.

Confidentiality: The information that is discovered while cracking the passwords should be reported immediately to the organization.

Scope Control: An ethical hacker can only do a bruteforce attempt only and not do anything that is beyond the scope because that would be illegal and they could use their certificate .

TASK 3: CEH 5 Phases of Mapping

- * Reconnaissance

-Since the ip address of the lab is known it can be used to determine how the different systems are connected.

- *Scanning

-The ip addresses can be used by the ethical hacker to scan if there are any open ports .

*Reporting

-Intrusion detection systems can be installed by the ethical hacker to detect any suspicious activities .

TASK 4 – Cyber Kill Chain Mapping

Reconnaissance (DNS footprinting/DNS Enumeration can be used to get the ip addresses of the lab.) → Weaponization(malware can be used to scan ports after discovering ip addresses to look for open ports can be prepared) → Delivery(The malware can be sent via email to one of the computers) → Exploitation(One of the users can click on the link sent via email to access this) → Installation(The malware can access the system of the lab.) → C2(The attacker gains access to the entire network through the malware that was created.) → Actions on Objectives(Since the attacker has control over network they can steal sensitive data and conduct ransomwares.)

PART 5 -CHAPTER 2 TASKS

TASK 5 – MITRE ATT&CK Mapping

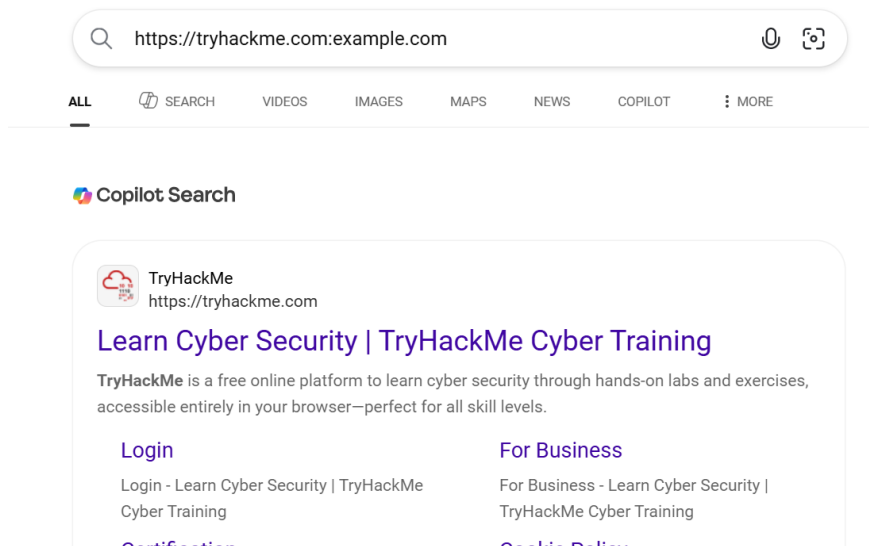
Reconnaissance Tactic → DNS enumeration → Gather information about the network.

Discovery Tactic → Ping sweep → Can be used to identify downtime and inactive systems .

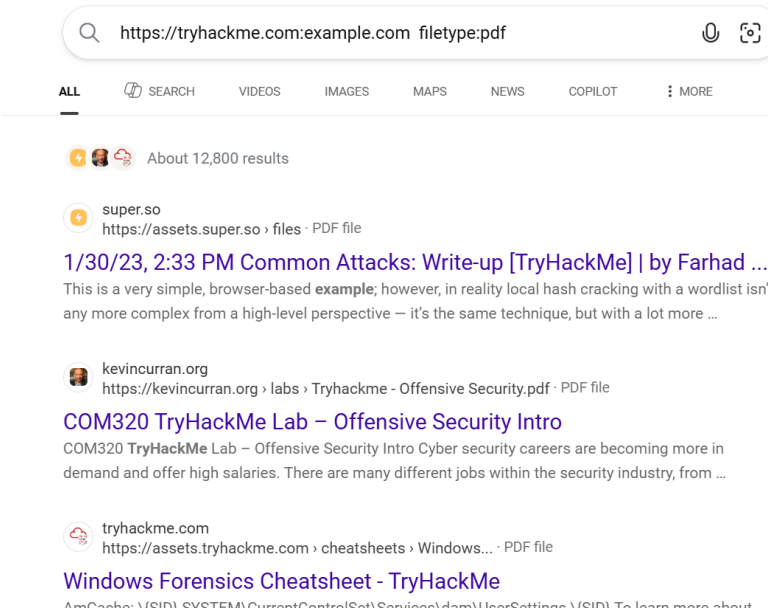
SECTION A

TASK 6 - PASSIVE FOOTPRINTING (GOOGLE OPERATORS)

Step 1: In this picture, the ethical hacker was gathering information about the target through search engines.



Step 2: The ethical hacker was looking for public documents that are displayed on search engines about the target.



*Reflection Question

-It is passive footprinting.

-It can help ethical hackers to come up with better ways to protect the data.

-It is dangerous because companies can access private data from their competitors and use it against them.

TASK 7 – WHOIS Enumeration

Step 1: The ethical hacker was using the whois command to gather the domain ownership and infrastructure information about the target. The ethical hacker was able to find the company that managed the domains, how old the domain is and the servers that are responsible for the dns resolution.

```
(kali㉿kali)-[~]  
$ whois example.com
```

```
$ whois example.com  
Domain Name: EXAMPLE.COM  
Registry Domain ID: 2336799_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.iana.org  
Registrar URL: http://res-dom.iana.org  
Updated Date: 2026-01-16T18:26:50Z  
Creation Date: 1995-08-14T04:00:00Z  
Registry Expiry Date: 2026-08-13T04:00:00Z  
Registrar: RESERVED-Internet Assigned Numbers Authority  
Registrar IANA ID: 376  
Registrar Abuse Contact Email:  
Registrar Abuse Contact Phone:  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: ELLIOTT.NS.CLOUDFLARE.COM  
Name Server: HERA.NS.CLOUDFLARE.COM  
DNSSEC: signedDelegation  
DNSSEC DS Data: 2371 13 2 C988FC423E3880EB8DD8A465E06CA230FE23E35B578D64E
```

TASK 8 – DNS Enumeration

Step1: The ethical hacker used the dig command to find the ipv4 address of the domain name example.com that is under the answer section.

```
(kali㉿kali)-[~]  
$ dig example.com
```

```
;; ANSWER SECTION:  
example.com.      149      IN      A       104.18.26.120  
example.com.      149      IN      A       104.18.27.120
```

Step 2: In the setup below, the ethical hacker used the dig command to discover the mail server domain.

```
(kali㉿kali)-[~]  
$ dig example.com MX
```

```
; <<>> DiG 9.20.15-2-Debian <<>> example.com MX  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 2257  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
;; QUESTION SECTION:  
example.com.                IN      MX  
  
;; ANSWER SECTION:  
example.com.                300     IN      MX      0 .  
  
;; Query time: 16 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)  
;; WHEN: Sat Feb 14 21:53:27 EST 2026  
;; MSG SIZE rcvd: 55
```

Step 3: In the figures below the ethical hacker used the dig example.com NS command to discover the server which translates the name of the target into an ip address

.

```
(kali㉿kali)-[~]  
$ dig example.com NS
```

```
; <<>> DiG 9.20.15-2-Debian <<>> example.com NS  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 62727  
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
;; QUESTION SECTION:  
example.com.                IN      NS  
  
;; ANSWER SECTION:  
example.com.                86400   IN      NS      hera.ns.cloudflare.com.  
example.com.                86400   IN      NS      elliottns.cloudflare.com.  
  
;; Query time: 12 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)  
;; WHEN: Sat Feb 14 21:57:50 EST 2026  
;; MSG SIZE rcvd: 95
```

Step 4: The ethical hacker used the dig TXT command to discover the txt records which contain the spf records .

```
(kali㉿kali)-[~]  
$ dig example.com TXT
```



```

; <<>> DiG 9.20.15-2-Debian <<>> example.com TXT
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 10010
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;example.com.                IN      TXT

;; ANSWER SECTION:
example.com.                300     IN      TXT      "_k2n1y4vw3qtb4skdx9e7dxt97q"
mmq9"
example.com.                300     IN      TXT      "v=spf1 -all"

;; Query time: 12 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
;; WHEN: Sat Feb 14 22:00:46 EST 2026
;; MSG SIZE rcvd: 109

```

TASK 9 – Subdomain Testing

Step 1: The ethical hacker used the nslookup command in the kali Linux ,to check if there are any sub domains and since there was a subdomain the ethical hacker was able to discover an ip address.

```

(kali@kali)-[~]
$ nslookup mail.example.com

Server:                192.168.1.1
Address:               192.168.1.1#53

```

TASK 10 – Website Technology Identification

Step1 : The ethical hacker ran a whatweb command to identify the technologies that were responsible for running web server .

```

(kali@kali)-[~]
$ whatweb http://example.com

http://example.com [200 OK] Allow[GET, HEAD], Country[UNITED STATES][US], HTML5, HTTPServer[cloudflare], IP[104.18.27.120], Title[Example Domain], UncommonHeaders[cf-cache-status,cf-ray]

```

TASK 12 – Active Host Discovery

Step 1: The ethical hacker used a nmap –sn command to scan this network to identify the number of devices which are active. The host up status simply means that the device is active and functioning properly.

```

Session  Actions  Edit  View  Help
(kali@kali)-[~]
$ nmap -sn 192.168.50.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-15 02:55 EST

```

```

Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-15 02:55 EST
Nmap scan report for 192.168.50.0
Host is up (0.40s latency).
Nmap scan report for 192.168.50.1
Host is up (0.0044s latency).
Nmap scan report for 192.168.50.2
Host is up (0.0043s latency).
Nmap scan report for 192.168.50.3
Host is up (0.0042s latency).
Nmap scan report for 192.168.50.4
Host is up (0.0045s latency).
Nmap scan report for 192.168.50.5
Host is up (0.21s latency).
Nmap scan report for 192.168.50.6
Host is up (0.21s latency).
Nmap scan report for 192.168.50.7
Host is up (0.21s latency).
Nmap scan report for 192.168.50.8
Host is up (0.20s latency).
Nmap scan report for 192.168.50.9
Host is up (0.20s latency).
Nmap scan report for 192.168.50.10
Host is up (0.20s latency).
Nmap scan report for 192.168.50.11
Host is up (0.40s latency).

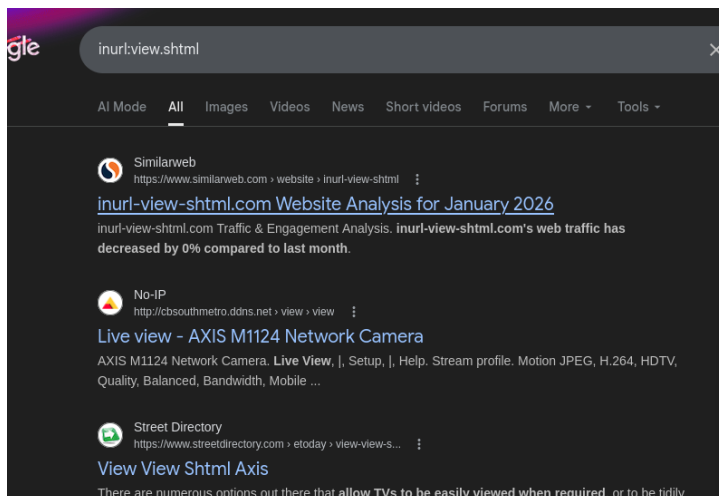
```

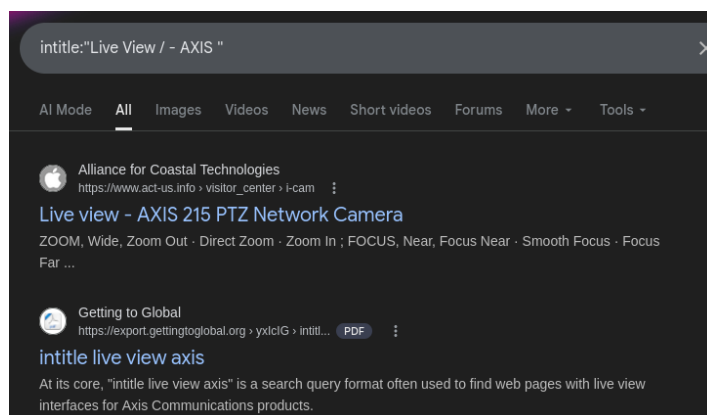
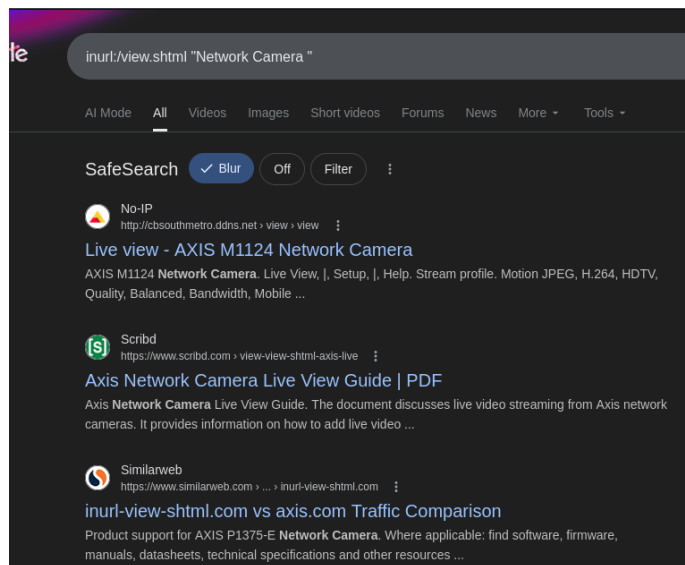
SECTION B: RECON LAB-ADDITIONAL OSINT TASKS

Live Cameras & Meeting Exposure Awareness

TASK 13 – OSINT: Public Live Camera Discovery (Pretoria Area)

Step 1: The ethical hacker used the different search engine operators to learn about how the organizations mistakenly expose their cameras due to the poor security practices that have been put in place. The risk that is created by the exposure of cameras is that it becomes very easier for malicious hackers to collect private information about the organizations. This is a passive reconnaissance because this information can be accessed on a public website.





TASK 14 – OSINT: Live Cameras in Pretoria (Geographic Intelligence)

Step1:

Is it legal?

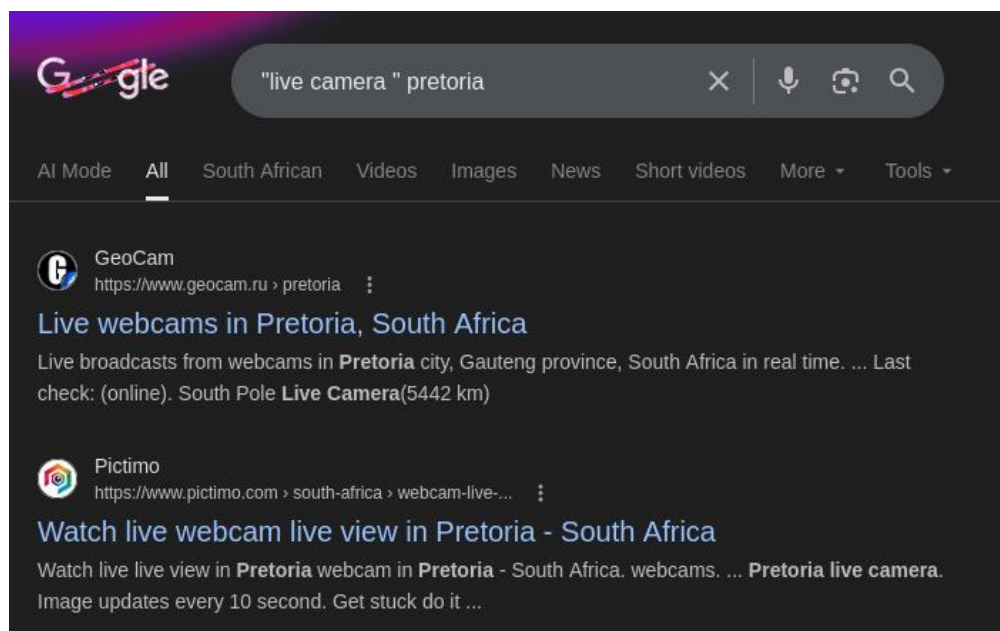
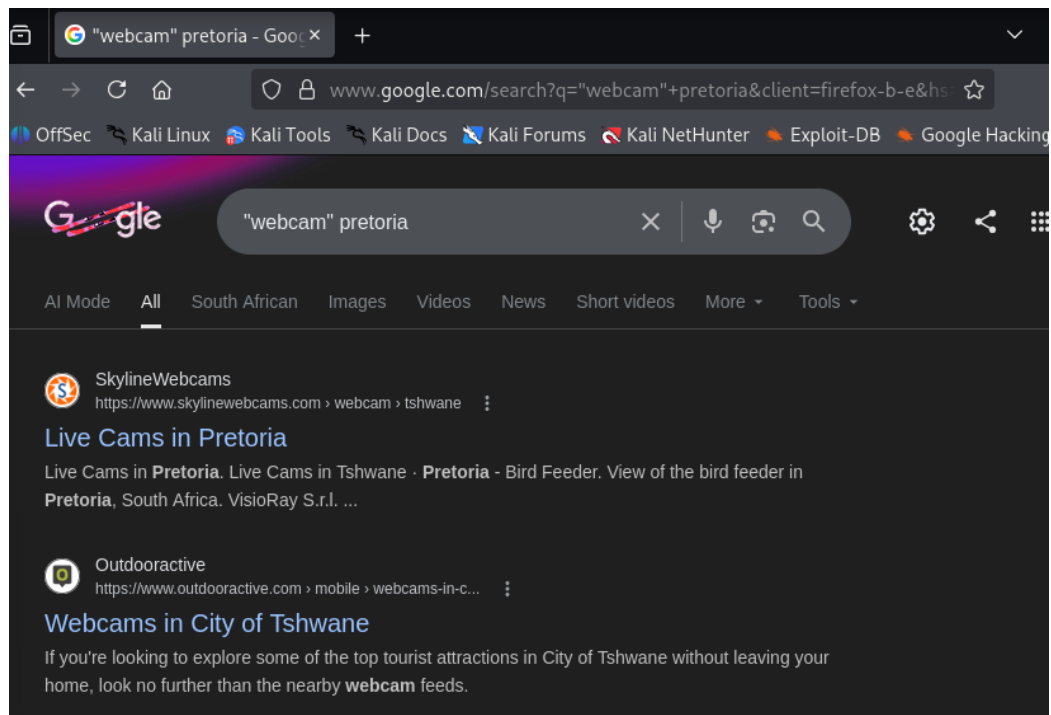
- This is not legal because it violates privacy laws, and anyone who violates such laws should be dealt with lawfully.

What makes it public?

-It is the fact that anyone that wants to access these cameras from the comfort of their homes.

How could this be abused?

-Some people may use these public cameras to commit crime by collecting private information without consent.



TASK 15 – OSINT: Public Google Meet / Teams Exposure Awareness



site:meet.google.com

ALL



SEARCH

IMAGES

VIDEOS

MAPS

NEWS

COPILOT

⋮ MORE



About 574 results



Google Meet

<https://meet.google.com>

[Google Meet - Online Video Calls, Meetings and ...](#)

Real-time meetings by Google. Using your browser, share your video, desktop, and presentations with teammates and customers.



"meet.google.com/" filetype:pdf

ALL



SEARCH

IMAGES

VIDEOS

MAPS

NEWS

COPILOT

⋮ MORE



About 132,000 results

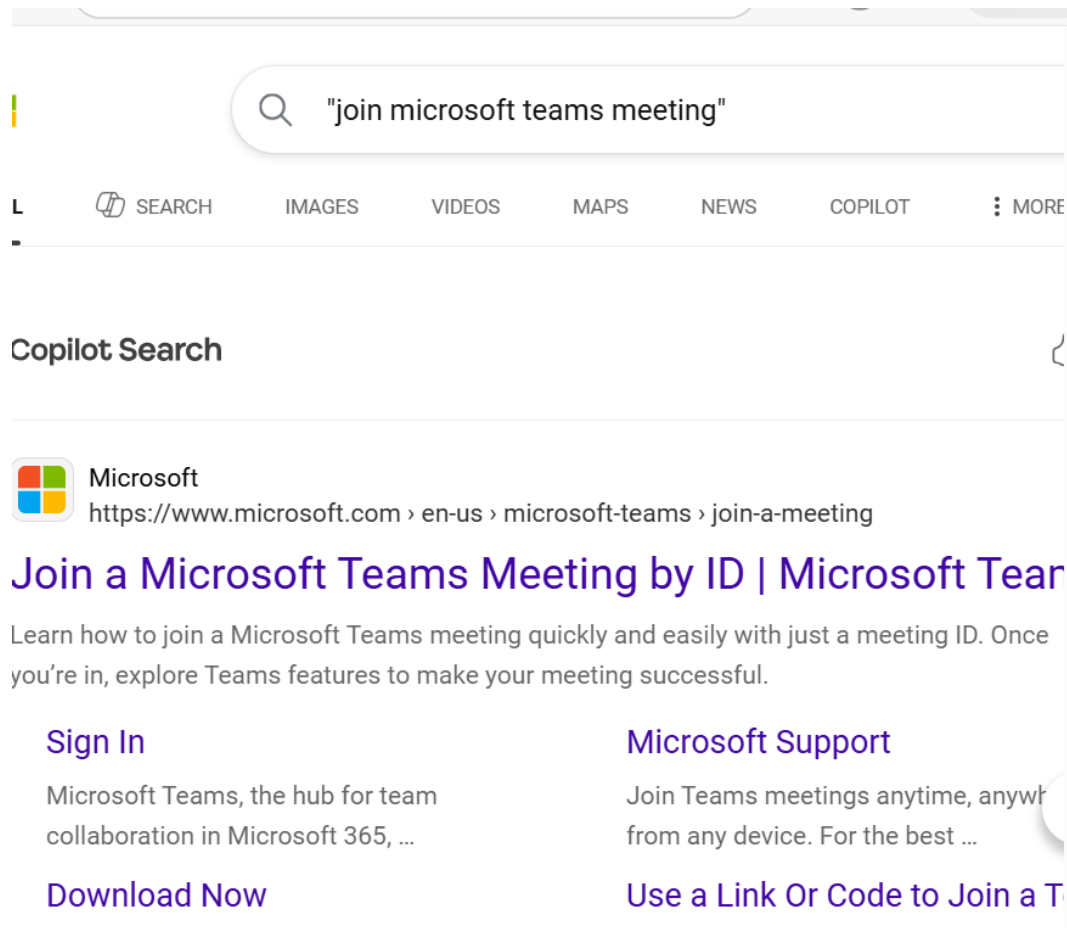


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[Google Meet User Guide - Literacy Minnesota](#)

Google Meet User Guide Google Meet User Guide This guide is designed to help users become familiar with the most common features and functions of Google Meet on a computer. With this guide, you can



Why is publishing meeting links dangerous?

It is dangerous because it increases the risk of sensitive data being exposed to external forces since anyone can join the meeting .

What mitigation should organizations implement?

-Do not share links publicly but only share them with the members of that particular organization.

-” Lock **Meetings Once Started**: After all invited participants have joined, lock the meeting to prevent late or unauthorized arrivals. “ (Burr B,2025)

TASK 16 – Ethical Reflection

1. Why is it illegal to access exposed cameras?

Authorization: It is illegal to access them because you do not have any permission from the user even if it is being exposed online.

Scope: An ethical hacker is supposed to stay within the scope, so access to these exposed cameras is not in the scope, and it can be illegal to do that.

Confidentiality: Cameras capture private spaces of individuals and business owners, so accessing them is a violation of confidentiality and privacy laws.

Response disclosure :If you find any exposed camera , you are supposed to report them and not take advantage of them.

2. Why is it illegal to join exposed meeting links?.

Authorization: It is illegal to access exposed meeting links because you are not invited, so you do not have permission.

Scope: An ethical hacker is suppose to stay on scope and in this case the scope was to document the exposure risk anything beyond the scope is illegal.

Confidentiality: In these meetings sensitive data is always shared and if someone joins the meeting without being invited, it violates confidentiality.

Responsible Disclosure: If one discovers such meetings, then they are supposed to report them to the organizations and not access them illegally.

3. What is the difference between:

Discovering exposure: discovering something that is available to the public, like exposed cameras for observation/research purposes without accessing it without authorized access.

Exploiting exposure: illegally accessing something like exposed cameras, meeting links and using them to access confidential information for your own gain.

4. Which CEH ethical principle applies here?

Authorization: Written authorization is mandatory.

Scope: Ethical hacking is strictly scope bound and out of scope activities are illegal even if no damage is caused.

Confidentiality: Confidential data discovered by hackers should not be shared, copied, or misused.

Responsible disclosure: Vulnerabilities must be reported accurately and responsibly.

References

[C. (2025). Ethical Considerations and Legal Aspects of Network Scanning with Nmap. [online] Siberoloji. Available at:] (<https://www.siberoloji.com/ethical-considerations-and-legal-aspects-of-network-scanning-with-nmap/#1-respect-for-privacy>)

[Accessed 14 Feb. 2026].

[Burr, B. (2025). 3 Common Security Risks in Virtual Conferencing Platforms (and How to Protect Your Meetings. [online] JetWebinar | All-in-One Platform for Engaging Virtual Events & Webinars. Available at:] (https://www.jetwebinar.com/blog/video-conferencing/3-common-security-risks-in-virtual-conferencing-platforms-and-how-to-protect-your-meetings/#Best_Practices_to_Prevent_Unauthorized_Access)

[Accessed 15 Feb. 2026].